Instructions

There are 2 possible themes to choose from. You are to **choose 1 theme to present your findings on,** and answer the questions for that theme. You do not need to justify your choice of theme.

You are to present your findings to the questions in a slide deck report consisting of **not more than 10 content slides** (excluding cover slide, appendices and references). The font used should be **Arial** and the minimum font size is **10pt**. There is no specific word limit for each slide.

Your first slide should be a cover slide consisting of your group name and group members. Appendices and references are permissible beyond the slide limit, but note that your analysis in these slides will not be graded. **It is important to keep your main insights and findings within the 10 content slides** (excluding the cover slide).

You are also to submit all relevant codes used for your analysis. All models and graphs should be reproducible, and you can provide instructions to the judging team on how to run your code. Code submissions will be strictly limited to that of **Python and R**, and no other languages will be entertained. It is recommended to provide comments for your codes. Note that a **20% penalty will be imposed if your code is not reproducible.**

Submit your completed slide deck report and codes in a **zip file** to NTU BAC Training Directors (abc-bac-training@e.ntu.edu.sg) by **23 October 2025, 2359 HRS**. Only one member needs to submit on behalf of the group. Late submissions will not be entertained.

Theme 1: Consumer Behaviour in E-Commerce

This dataset provides a comprehensive collection of consumer behaviour data that can be used for various market research and statistical analyses. It includes information on purchasing patterns, demographics, product preferences, customer satisfaction, and more, making it ideal for market segmentation, predictive modelling, and understanding customer decision-making processes.

The dataset is designed to help researchers, data scientists, and marketers gain insights into consumer purchasing behaviour across a wide range of categories. By analysing this dataset, users can identify key trends, segment customers, and make data-driven decisions to improve product offerings, marketing strategies, and customer engagement.

The link to download the dataset is as follows:
https://drive.google.com/file/d/1LayWoxeoGPrgZ9jFE9rSH8L3QQdQQCvo/view?usp=drive_link

You may view the Kaggle link for more information on the dataset description and features:
https://www.kaggle.com/datasets/salahuddinahmedshuvo/ecommerce-consumer-behavior-analysis-data

Questions to Answer:

(a) Quantitative Questions (1-2 Slides suggested, working shown in code as well):

1. Calculate the median consumer age.
2. Identify the number of distinct purchase categories. Count the number of entries for each category.
3. Calculate the mean purchase amount.
4. Identify the number of distinct purchase channels. Count the number of entries for each purchase channel.
5. Complete the following table:

Variable	Variable Type (Continuous/Categorical)
Age	Continuous
Gender	Categorical
Brand Loyalty	
Product Rating	
Customer Satisfaction	
Frequency of Purchase	
Time to Decision	

(b) Qualitative Questions (8-9 Slides suggested):

1. Briefly describe your data cleaning process.
2. Formulate a suitable business problem to be addressed based on the dataset. You may include data or references to support your formulation.
3. As part of exploratory data analysis, utilise appropriate charts/graphs to gather insights from the cleaned data. These charts/graphs should be relevant to your formulated business problem.
4. Utilise up to 3 appropriate analytical/predictive models to address your formulated business problem. Describe and explain the key technical features and parameters of each analytical model, and explain its relevance to solving your business problem.
5. Based on your analytical/predictive models, what are the key business findings that can be observed? How can e-commerce companies use these findings in their business operations?
6. What personalised strategies can be recommended for e-commerce companies to improve customer sales?

Theme 2: Customer Retention & Telecom Analytics

JB Link is a small fictional telecom company in California that provides phone and internet services and has grown rapidly in six years by expanding infrastructure to underserved regions. Despite strong new customer acquisition in the previous quarter, customer retention is a major issue.

To address this, JB Link has formed a cross-functional task force, including a new Data Science team, to develop a customer retention strategy. The data team's objectives are to:

- Analyse data to understand drivers of churn
- Build predictive models to identify customers at risk
- Recommend personalized actions to retain customers

Executives aim to reduce churn because acquiring new customers costs up to five times more than retaining existing ones, making this project crucial for cost savings and company growth.

The link to download the dataset is as follows:
https://drive.google.com/file/d/16wDTUrkqIbGsirFy4E0xjoqtys3Q7FYr/view?usp=drive_link

You may view more information on the dataset in the corresponding Kaggle Link:
https://www.kaggle.com/datasets/johnflag/jb-link-telco-customer-churn/data

Questions to Answer:

(a) Quantitative Questions (1-2 Slides suggested, working shown in code as well)

1. What is the median tenure of churned customers and the median tenure of non-churned customers?
2. What are the top 5 countries with the highest churn rates? What is the most common reason for churn for the 5 countries, respectively?
3. Which customer segments have the highest churn rates based on demographics, location, or service usage?
4. What are the distributions of the payment methods adopted by customers?
5. Complete the following table:

Variable	Variable Type (Continuous/Categorical)
Tenure in Months	Continuous
Payment Method	Categorical
Monthly Charge	
Gender	
Age	

Churn Value	
Customer Satisfaction	

(b) Qualitative Questions (8-9 Slides suggested):

1. Briefly describe your data cleaning process.
2. Formulate a suitable business problem to be addressed based on your dataset. You may include data or references to support your formulation.
3. As part of exploratory data analysis, utilise appropriate charts/graphs to gather insights from the cleaned data. These charts/graphs should be relevant to your formulated business problem.
4. Utilise up to 3 appropriate analytical/predictive models to address your formulated business problem. Describe and explain the key technical features and parameters of each analytical model, and explain its relevance to solving your business problem.
5. Based on your analytical/predictive models, what are the key business findings that can be observed? How can JB Link use these findings in its business operations?
6. What strategies can be recommended for JB Link to reduce customer churn?

Judging Criteria for Preliminary Round

Criteria	Weight	Excellent	Good	Basic	Poor
1. Depth of Analysis	25%	- Demonstrates deep analytical thinking by uncovering meaningful patterns, relationships, or anomalies across multiple variables - Clear reasoning supports insights - Business problem is excellently substantiated	- Provides reasonable analysis with some variable connections - Lacks depth or misses key relationships - Business problem is well substantiated	- Surface-level analysis with basic observations - Limited exploration beyond obvious trends - Business problem is poorly substantiated	- Little to no meaningful analysis - Mostly descriptive or repetitive - No substantiation of business problem
2. Use of Data & Methodological Rigour	20%	- Uses data effectively with relevant statistics, visualisations, and analytical models - Justification and explanation of modelling process is strong - Methodology is sound, justified, and correctly applied	- Makes some use of data, but methods may be simplistic or partially justified	- Minimal data usage; analysis relies on general statements or assumptions	- Does not reference data meaningfully - Methodology is unclear or inappropriate
3. Quantitative Accuracy	15%	- Calculations are accurate, assumptions are logical, and interpretations are correct	- Minor errors in calculation or interpretation that do not significantly affect conclusions	- The response contains significant errors in calculations or interpretation	- The response is completely incorrect in both calculations and interpretation

4. Creativity & Insight	15%	- Provides novel perspectives or explores the dataset in innovative ways	- Shows some originality but remains close to standard approaches	- Standard, expected analysis with little creative exploration	- No creative or original approach
5. Structure, Clarity & Communication	15%	- Submission is well-structured, visually clear (charts/tables), and arguments flow logically - Insights are easy to follow	- Generally clear, with minor issues in structure or visualisation	- Somewhat disorganised or unclear, making interpretation harder	- Disorganised and unclear - Hard to follow logic
6. Relevance to Guiding Questions	10%	- Addresses all guiding questions comprehensively, demonstrating clear understanding of the problem context	- Addresses most questions but misses some depth or clarity	- Addresses only part of the questions superficially	- Ignores or misunderstands key guiding questions

Penalty:

- **20% penalty** to be imposed if **code cannot be reproduced**

FAQs

1. **How should I structure the slide deck report?**

Answer: Your slide deck report should be clear, concise and structured to highlight your main insights within 10 content slides. The first slide should be a cover slide with your group name and group members, and it is excluded from the slide count. It is recommended that you follow the guiding questions provided and plan your slides accordingly. Any additional information can be included in the appendix, which is beyond the slide limit.

2. **What if I exceed 10 content slides for the slide deck report?**

Answer: The extra slides will not be considered for grading. Only appendices and references will be allowed beyond the 10 content slide limit.

3. **How are the quantitative questions graded?**

Answer: Accuracy will be graded under Criteria 3 (Quantitative Accuracy). Wrong or incorrect calculations will result in penalties. Refer to the judging criteria for more details

4. **How are the qualitative questions graded?**

Answer: Refer to the judging criteria for more details.

5. **Are there any penalties if my codes cannot be run/if my coding results cannot be reproduced?**

Answer: An automatic 20% penalty will be imposed if your codes cannot be run on the end of the judging team.

6. **Is there any limit to how many slides I can have for appendices or references?**

Answer: There is no limit. However, appendices and references should be relevant to your findings.

7. **What happens if I submit my files late?**

Answer: Late submissions will not be entertained and will lead to automatic disqualification.

8. **Am I allowed to modify the dataset?**

Answer: You may not add columns or rows to the data. You are only allowed to clean the data..

9. **Who do I contact if I have any further questions?**

Answer: You may send an email to NTU BAC Training Directors (abc-bac-training@e.ntu.edu.sg). Your query will be responded to as soon as possible.